BrandPulse
Analytics

Analyzing sentiment, topics and emojis in brand social media posts

Team Assignment 2 - May 2026

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Key Findings



Sentiment drives engagement, but only for some brands

A **positive tone significantly boosts likes** for TechNest (strongest and most significant effect) and FitPulse, but has no statistically significant impact for LumoraBeauty suggesting tone matters more in tech and fitness than in beauty.



Emojis are the most consistent driver of engagement across all brands

Emoji count is the only predictor that significantly increases likes for all three brands (LumoraBeauty, FitPulse, TechNest), though the effect is strongest for LumoraBeauty.



Brand content is predominantly informational, not promotional

The three brands rely more oninformative content than entertainment or promotional. 68.5% of all posts were classified as Informative, 24% as Promotional, and only 7.5% as Entertaining



Entertaining content is a high-reward strategy, but only for FitPulse

Entertaining posts generate higher number of likes for FitPulse while this content type has no significant effect on Lumora or TechNest.

Engagement drivers are specific for each brand.

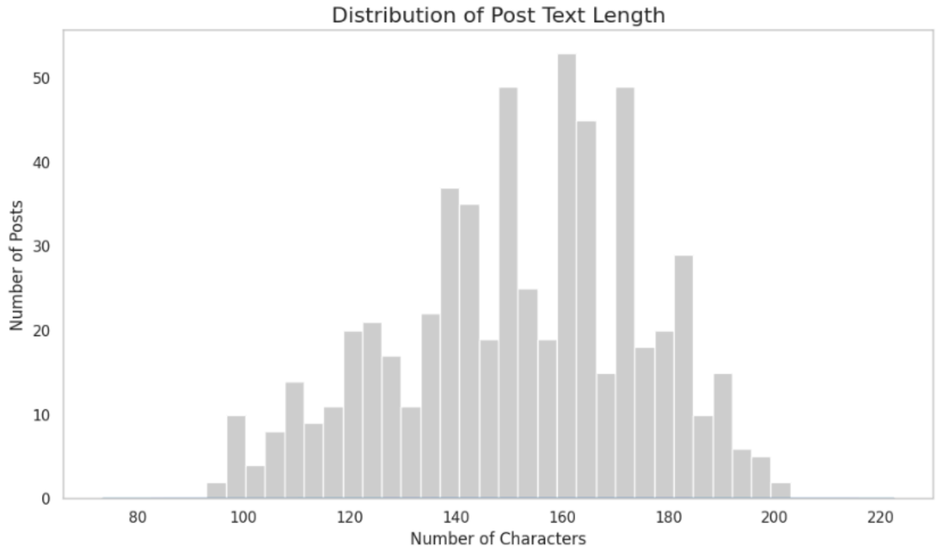
Data Quality & Structure

- Our dataset comprises **600 social media posts** equally distributed across three brands: **LumoraBeauty, FitPulse, and TechNest** (200 posts each).
- **9 variables** including Post_Text, Brand, Handle, Total_Followers, Total_Posts_LastYear, Num_Likes, Num_Comments, and Num_Reposts.
- **Data quality verified:** 0 missing values, 0 duplicates, UTF-8 encoding confirmed for accurate emoji extraction.

Quality Checks

Missing values	**PASS**
Duplicate rows	**PASS**
Duplicate Post_IDs	**PASS**
UTF-8 emoji encoding	**PASS**
Variable types verified	**PASS**

Post text lengh distribution

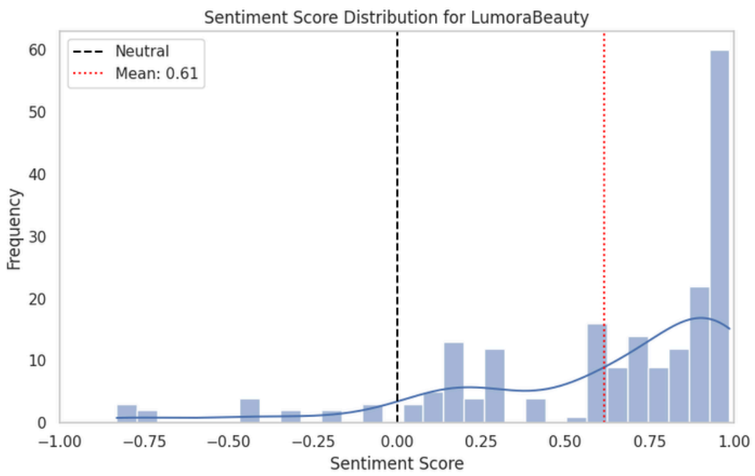


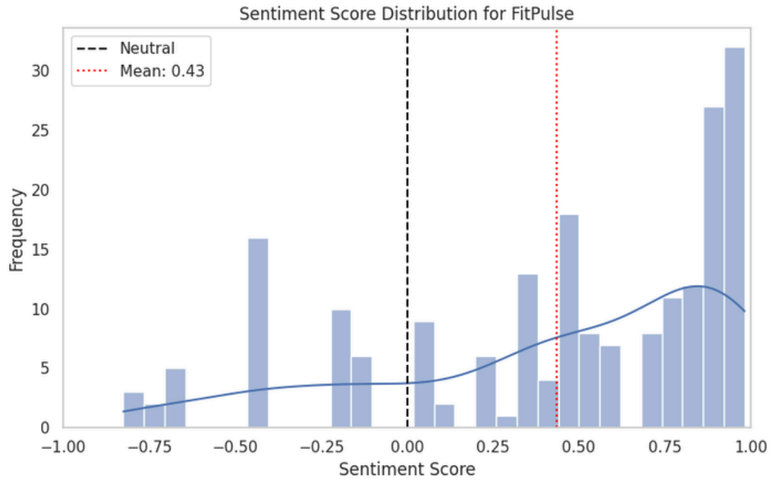
Sentiment Analysis: How positive are the posts?

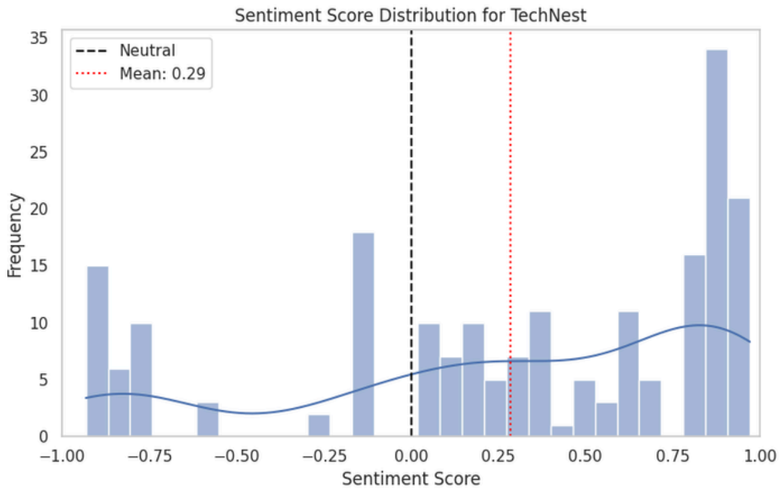
Model: Hugging Face Transformers *cardiffnlp/twitter-roberta-base-sentiment-latest*

Our sentiment analysis reveals distinct brand personalities:

- **Lumora Beauty** achieves the highest positive sentiment scores, reflecting their aspirational messaging.
- **FitPulse** shows more neutral and slightly negative
- **TechNest** shows more neutral and varied sentiment, typical of informative tech content.







*-**Positive (FirPulse):** Hot take: the best time to work out is whenever you will actually do it 😅 Morning, lunch, evening. What time works for you? Tell us 👇*
*-**Negative (TechNest):** Cheap USB-C cables can damage your devices 🚨 Poorly manufactured cables may deliver incorrect voltage. Always buy cables that meet USB-IF certification standards ⚠️*
*-**Neutral (LumoraBrand):** Skincare personality test 🧴 Are you a 3-step minimalist or a full 10-step devotee? No wrong answers 👇*

Topic classification: What are Brands posting about?

🔍 Informative:

FitPulse is the most informative brand with **72.5%** of its posts being more educational and advice oriented.

Example: *"Watching your form improve over months of consistent training hits different 🎯💪 It is not just the muscles. It is the discipline, the belief in yourself. That is the real gain 🙌🔥"*

🎉 Entertaining:

TechNest is the most entertaining brand with **10.5%** posts relating to its products or routines.

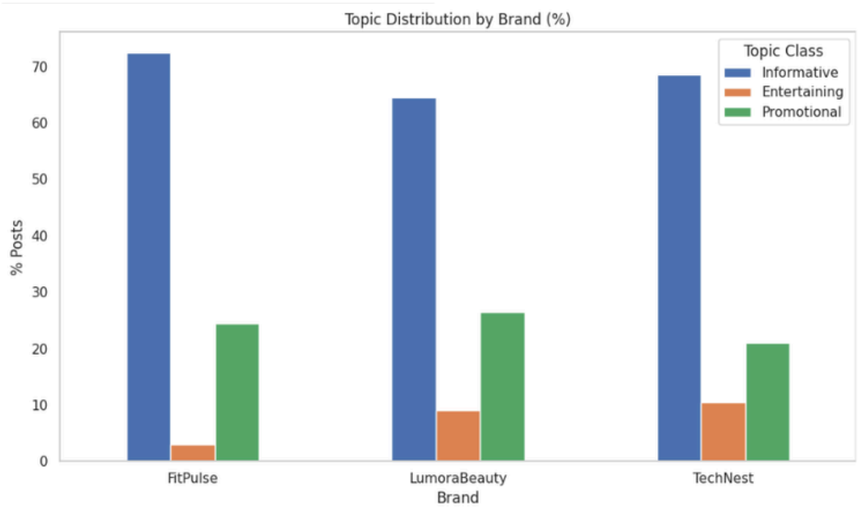
Example: *"The most powerful productivity hack is closing 47 browser tabs 😂💻 Currently sitting at ___ tabs. Fill in the blank 👇 No judgement (there is definitely some judgement)"*

🏷️ Promotional:

LumoraBeauty is most promotional, dominates with community oriented posts **26.5%**.

Example: *"Glow-up season is every season with Lumora 🌸✨ Show us your before and after. We love celebrating YOUR progress 💕🙌"*

Brand	Informative	Entertaining	Promotional
FitPlus	145	6	49
LumoraBeauty	129	18	53
TechNest	137	21	42
Total	411	45	144



Emoji Analysis

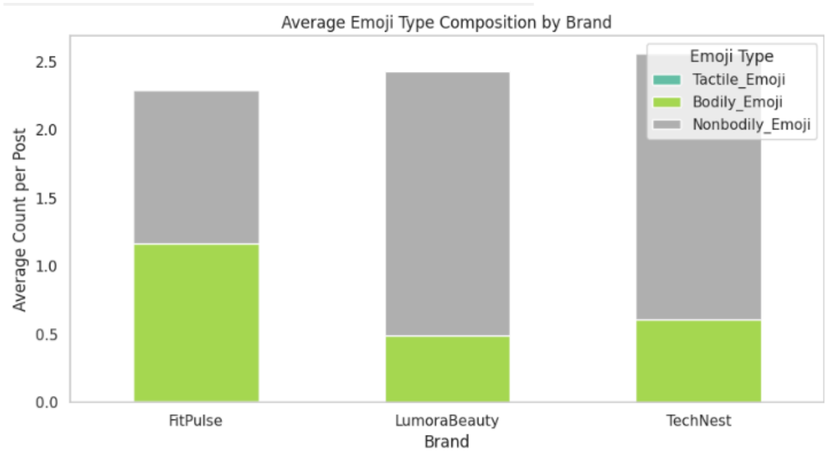
Nonbodily emojis dominate the emoji mix, followed by Bodily emojis. Tactile emojis are absent.

- **FitPulse:** Uses balanced proportion between bodily and nonbodily emojis. This is the only brand in which bodily emojis are dominant showing a consistency with its brand identity
  Most used emojis: 💪🔥 👇

- **LumoraBeauty:** Nonbodily emojis are dominant while bodily emojis usage is the least across all brands. It aligns with its aspiratonal identity.
  Most used emojis: ✨💛🌸

- **TechNest:** Shows the highest proportion of Nonbodily emojis including tech oriented emojis. It is the brand with highest emoji average (2.56)
  Most used emojis: ⚡💻👇🖥️

Brand	Avg.	Tactile emoji	Bodily emoji	Nobodily emoji
FitPlus	2.30	0.01	1.16	1.12
LumoraBeauty	2.44	0	0.49	1.94
TechNest	2.56	0	0.61	1.96

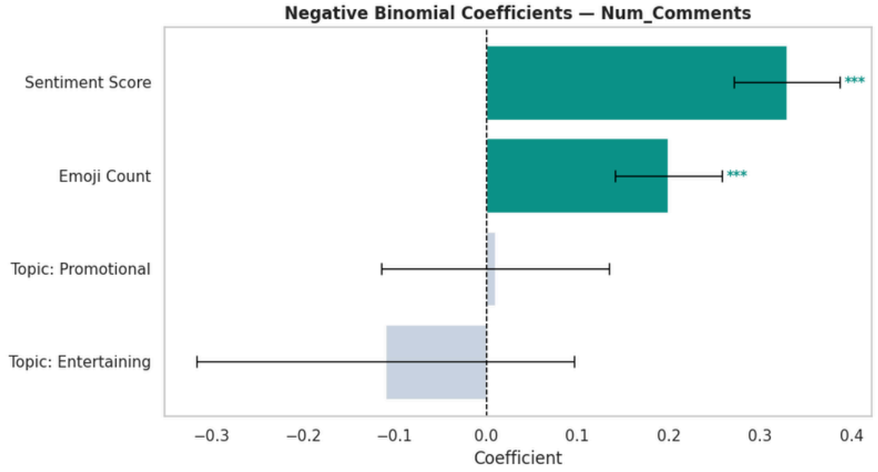


- *There is a moderate positive correlation between emoji count and sentiment score (r = 0.37). Posts with more emojis tend to score higher on positive sentiment, suggesting that emoji usage and positive tone co-occur in brand content though the relationship is not strong enough to imply causation*

Drivers of engagement: What makes posts perform?

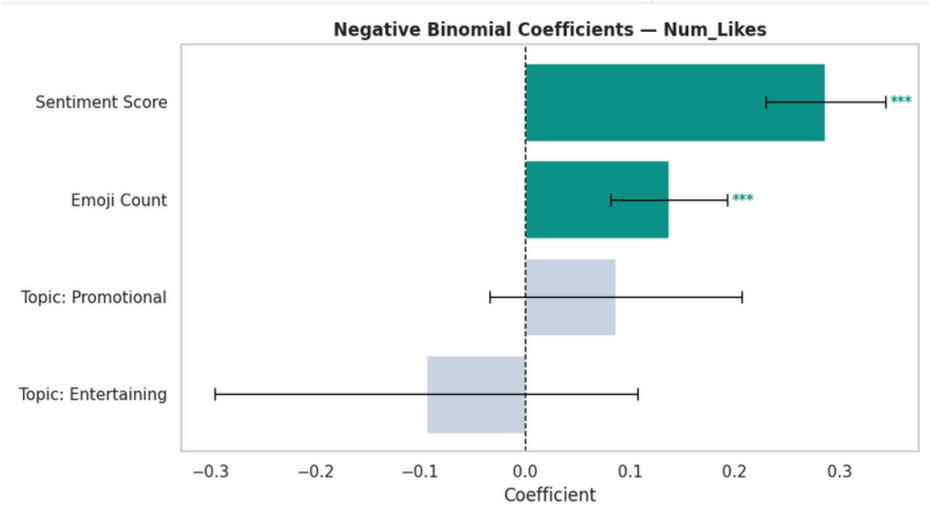
Number of Comments

- **Sentiment Score** is the strongest driver ***positive tone drives conversation even more than likes***
- **Emoji Count** is significant, ***emojis drive comments consistently***
- **Topic Promotional** coefficient is nearly zero and not significant
- **Topic Entertaining** is negative and not significant



Number of Likes

- **Sentiment Score** is the strongest ***posts with a more positive tone get significantly more likes***
- **Emoji Count** is the second significant driver ***more emojis consistently drive more likes***
- **Topic Promotional** is positive but not significant. ***Promotional content does not reliably drive likes***
- **Topic Entertaining** is negativeand not significant



Brand differences: Do Engagement Drivers Differ Across Brands?

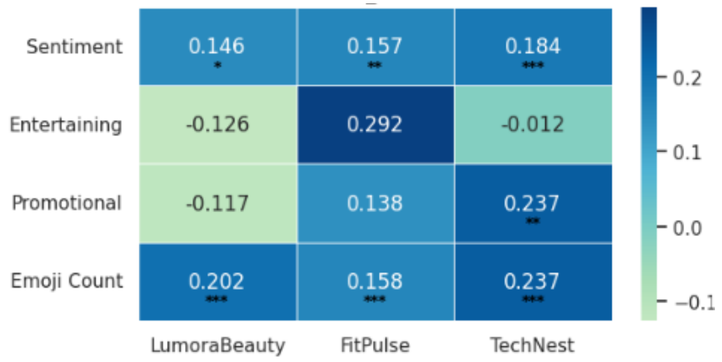
Number of Comments

- Emoji count and Sentiment score is the main driver across all brands
- Sentiment score is strongest for LumoraBeauty, how positively a post is perceived drives engagement more on the comments side than likes.
- Emoji count is strongest for TechNest, considering they also use the most emojis on average (2.56 per post), Increasing emoji use or using them more strategically could meaningfully boost both likes and comments.



Number of Likes

- Emoji count is the main driver across all brands
- Sentiment score and emoji count are the strongest predictors for LumoraBeauty and FitPulse number of likes
- Entertainment Topic for FitPulse is the largest coefficient but is not significant
- For TechNest, likes are driven by sentiment, emoji usage, and promotional content



- Sentiment is consistently stronger for comments than likes across all brands
- Sentiment Score for LumoraBeauty is not highly significant for Likes but significant for Comments, sentiment matters for conversation but not for likes
- Emoji Count is more balanced, significant for both, for comments and likes

Recommendations

Universal Engagement Strategies

Use positive, engaging content with strategic emoji use to maximize likes and comments.

Lumora Beauty Focus

Leverage heavy emoji style and aspirational tone; prioritize positive sentiment posts.

FitPulse Content Strategy

Prioritize entertaining content formats over promotional ones; humor and fun convert directly into likes.

TechNest Approach

Combine promotional messaging with upbeat sentiment and rich emoji use to maximise both metrics.

Thank You.



Appendix and variable dictionary (1/2)

- **Google colab notebook:** https://colab.research.google.com/drive/1J43lAIZE_TRTcHbPC0bIxBGA6k_n2VsJ?usp=sharing

Variable	Type	Description
Post_ID	int	Unique post identifier (1–600)
Brand	string	LumoraBeauty / FitPulse / TechNest
Handle	string	Social media handle (@handle)
Post_Text	string	Full text of post including emojis (UTF-8)
Total_Followers	int	Total follower count at data collection
Total_Posts_LastYear	int	Posts published in the 12 months prior
Num_Likes	int	Total likes received on the post
Num_Comments	int	Total comments/replies received
Num_Reposts	int	Total reposts/shares
Sentiment_Score	float	HuggingFace RoBERTa: Positive prob − Negative prob (−1 to +1)
VADER_Compound	float	VADER compound score (−1 to +1), robustness check
HF_Sentiment_Label	string	Predicted label: Positive / Neutral / Negative

Appendix and variable dictionary (2/2)

Variable	Type	Description
Topic_Class	string	Zero-shot: Informative / Entertaining / Promotional
Emoji_Count	int	Total number of emojis per post (PARA)
Bodily_Emoji	int	Count of bodily emojis (gestures, body parts)
Nonbodily_Emoji	int	Count of non-bodily emojis (objects, nature)
Tactile_Emoji	int	Count of tactile emojis (touch/contact)
TPL_Index	int	Total Paralanguage Index — expressiveness score
Emoji_Index	int	PARA emoji index score per post
Pitch / Rhythm / Stress	int	Vocal paralanguage markers
Emphasis / Tempo / Volume	int	Vocal paralanguage markers
Censorship / Spelling / Alternant	int	Textual paralanguage markers
Differentiator / Alphahaptics / Alphakinesics /	int	Textual paralanguage markers
Tactile / Bodily / Nonbodily Emoticon	int	Emoticon type counts per post